

Project | Sustainability Impact Analysis for Intel



INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO₂ emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

NOTE: The dataset you are working with is designed for the Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

– Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"

- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

– Task 1: Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

- A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant data is included in your result.

(paste your query below 📌)

```
SELECT
  a.*,
  b.*
FROM intel.device_data AS a
```

```
FULL JOIN intel.impact_data AS b
ON a.device_id = b.device_id
```

- B. To your joined dataset, add a new column called `device_age` calculated by subtracting the `model_year` from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 📌)

```
SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
ON a.device_id = b.device_id
```

- C. Order your joined data by `model_year` (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 📌)

All of the devices when we ordered them this way on the joined data were at least 5 years old. This may indicate more older devices, but grouping the data by year would give us more meaningful insights. More older devices being repurposed could indicate that repurposing older devices is a priority, as they may just be thrown out and wasted if they don't find a new use soon.

- D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO₂ reductions more effectively than using individual ages. Use a `CASE`

WHEN clause to add one more column, called `device_age_bucket`, to your data, that is based on the `device_age`:

- WHEN the `device_age` is less than or equal to 3, `device_age_bucket` should be “newer”
- WHEN the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be “mid-age”
- WHEN the `device_age` is greater than 6, `device_age_bucket` should be “older”

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as “mid-age”.

(paste your query below 📌)

```
SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age,
  CASE
    WHEN (2024 - a.model_year) <= 3
    THEN 'newer'
    WHEN (2024 - a.model_year) <= 6
    THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
  ON a.device_id = b.device_id
ORDER BY a.model_year ASC
```

– Task 2: Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the `WITH` keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. What is the total number of devices Intel repurposed in 2024? Remember, the dataset is already representing all devices repurposed in 2024!

(write your **answer** below 📌)

There are 601,740 devices being repurposed.

- A. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO₂ emissions saved (in tons) from repurposed devices.

(paste your query below 📌)

```
WITH joined_data AS (SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age,
  CASE
    WHEN (2024 - a.model_year) <= 3
    THEN 'newer'
    WHEN (2024 - a.model_year) <= 6
    THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
  ON a.device_id = b.device_id
ORDER BY a.model_year ASC)

SELECT
  COUNT(*) AS total_devices,
  AVG(device_age) AS avg_device_age,
  AVG(energy_savings_yr) AS avg_energy_savings,
  SUM(co2_saved_kg_yr) / 1000 AS tons_co2_saved
FROM joined_data
```

- B. Now that you have calculated the average estimated energy savings (kWh) and CO₂ emissions saved (tons), use ChatGPT to help put these numbers into perspective.



Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO₂ emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO₂ produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO₂ emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 🖊)

I thought it was really cool that if intel repurposed 100,000 devices, it saves enough electricity to power around 240 homes for one year. We know that Intel repurposed just over 6 times this amount, so in one year, they save enough electricity to power around 1,440 houses. In terms of CO₂, the average car emits around 4.6 tons every year. Intel's CO₂ savings are the equivalent of taking around 1,470 cars off the road for a year. Both of these figures highlight how small actions can quickly add up and make a real impact.

– Task 3: Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO₂ reductions. These insights will help us determine which categories of devices contribute the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

- A. Write a query that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), grouped by device_type.

(paste your query below 🖊)

```

WITH joined_data AS (SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age,
  CASE
    WHEN (2024 - a.model_year) <= 3
    THEN 'newer'
    WHEN (2024 - a.model_year) <= 6
    THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
  ON a.device_id = b.device_id
ORDER BY a.model_year ASC)

SELECT
  device_type,
  COUNT(*) AS total_devices,
  AVG(energy_savings_yr) AS avg_energy_savings,
  AVG(co2_saved_kg_yr) / 1000 AS avgtons_co2_saved
FROM joined_data
GROUP BY device_type

```

- B.** Based on the results, which device type contributes the most to energy savings and CO₂ reduction? Why might that be the case?

(write your **answer** below 🖊)

Laptops barely edge out desktops for average energy savings and CO₂ savings. The numbers are very similar though. However, there are many more laptops, a little over 400,000, compared to slightly less than 200,000 desktops. Based on the fact that the average savings per device are similar, laptops contribute the most to the savings because there are more than double the number of laptops vs. desktops.

- C. Write a query that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), now grouped by `device_age_bucket`.

(paste your query below 📌)

```
WITH joined_data AS (SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age,
  CASE
    WHEN (2024 - a.model_year) <= 3
    THEN 'newer'
    WHEN (2024 - a.model_year) <= 6
    THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
  ON a.device_id = b.device_id
ORDER BY a.model_year ASC)

SELECT
  device_age_bucket,
  COUNT(*) AS total_devices,
  AVG(energy_savings_yr) AS avg_energy_savings,
  AVG(co2_saved_kg_yr) / 1000 AS avgtons_co2_saved
FROM joined_data
GROUP BY device_age_bucket
```

- D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 📌)

The older a device is, the more energy it will save on average. Even though there are less older devices, the average energy savings is a lot higher on these. Finding the priority on which devices to repurpose first will need more analysis, but we know for sure that the newer a device is, the less energy we save by repurposing it.

- E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO2 emissions saved (in tons), now grouped by region.

(paste your query below 📌)

```
WITH joined_data AS (SELECT
  a.*,
  b.*,
  2024 - a.model_year AS device_age,
  CASE
    WHEN (2024 - a.model_year) <= 3
    THEN 'newer'
    WHEN (2024 - a.model_year) <= 6
    THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM intel.device_data AS a
FULL JOIN intel.impact_data AS b
  ON a.device_id = b.device_id
ORDER BY a.model_year ASC)

SELECT
  region,
  COUNT(*) AS total_devices,
  AVG(energy_savings_yr) AS avg_energy_savings,
  AVG(co2_saved_kg_yr) / 1000 AS avgtons_co2_saved
FROM joined_data
GROUP BY region
```

- F. How does the carbon intensity of electricity in each region impact the total CO₂ savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 🖊)

Asia saves the most CO₂ on average, despite all of the energy savings being extremely close for each region. This is likely because Asia's electricity production is much more carbon-intensive. Saving energy saves a lot of CO₂ here. This is a different story in Europe, where the average CO₂ savings are much lower. This is likely the result of more electricity in Europe coming from cleaner sources, such as wind or solar. Saving energy in Europe does not have as much of an impact on emissions because their energy is cleaner overall. North America sits between the two, implying that energy production is cleaner than Asia but pollutes more than in Europe. After this analysis, we can see that repurposing in Asia will lead to higher environmental benefits and their region should be the priority for repurposing.

– Task 4: Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO₂ reductions while ensuring the most effective use of resources.

- A. Based on your analysis of the repurposed devices (including energy savings, CO₂ emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 🖊)

-Older devices save more energy when repurposed. They should be the priority, as they are likely to be thrown away if they are not quickly

repurposed. Repurposing 10,000 old devices will have a bigger impact than repurposing 10,000 newer devices.

-The target region should be Asia. Repurposing devices saves much more CO₂ here than in other regions, as their energy production is more carbon-intensive.

-The focus should be on laptops. Because there are way more laptops than desktops and their energy and CO₂ savings are very similar, laptops should be the priority. Bonus points if they are older laptops as well.

-Participation in the program matters. The more devices we see being entered into the repurposing program, the more the data will change. Having other types of devices or more of the ones we already analyzed could possibly improve our results.

- B. Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO₂ reductions.

(write your **answer** below 🙌)

For Intel, I would recommend focusing on old laptops and desktops first. If they put most of their resources into that, they will be able to fix the old devices pretty quickly and move onto the newer ones later because there are less old devices. I would also try to focus mostly on the Asian market, as this will produce the greatest net positive environmental benefits. To summarize, I would recommend focusing on old laptops and desktops in Asia, at least at first. Once those are fixed, I would then focus on old devices in North America. I would continue this pattern all the way until I got to fixing new devices in Europe. I would also possibly provide incentives or ways to get more people to participate in the repurposing program.

- C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 📌)

ChatGPT highlighted the importance that participation in the program matters the most. If there aren't devices to repurpose, the insights don't matter because we can't put them into action. ChatGPT also emphasized how geographic location plays a large role. Focusing on Asia will bring the highest overall impact on the planet.